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ABSTRACT

This study investigates magnetization relaxation effects in ferrofluids using a coupled molecular dynamics and Lattice Boltzmann simulation framework. The model incorporates dipole-dipole and short-range (Lennard-Jones) interactions between particles, couples hydrodynamic interactions, and employs a bidisperse model for particle distribution. The magnetization component along the magnetic field direction, M_x^*, increases and saturates with the Langevin parameter α_L. However, larger shear rates γ^* or pressure differences P^* (beyond a threshold) slow this saturation due to the greater difficulty larger particles face in aligning with the magnetic field. The off-field magnetization component M_y^*, arising from relaxation, exhibits a non-monotonic dependence on α_L, first increasing and then decreasing under shear and pressure difference. This behavior is attributed to the evolving chain-like microstructures within the ferrofluid at different field intensities. Higher particle concentrations φ_m or dipole coupling parameters λ lead to larger maximum values of M_y^*. The competition between the magnetic field and shear flow, along with shear-rate-dependent chain size distributions, leads to a larger γ^* requires a stronger magnetic field for M_y^* to peak, and the peak value itself increases with γ^*. The pressure difference P^* influences magnetization by modifying the local shear rate distribution, with its hydrodynamic effects becoming significant only after P^* exceeds a critical value.

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I. INTRODUCTION

A ferrofluid is a colloidal solution consisting of nanoscale ferromagnetic or ferrimagnetic solid particles coated with surfactants and suspended in a carrier fluid.[1,2] Due to the combination of magnetic properties and fluidity, ferrofluids are widely used in engineering and technology fields, including sealing,[3,4] vibration damping,[5] lubrication,[6] sensing,[7] and biomedicine.[8] Under the influence of a magnetic field, a ferrofluid becomes magnetized. The equilibrium magnetization in a static state is determined by the balance between two factors: the magnetic field energy and the thermal kinetic energy of the particles within the ferrofluid. This relationship is typically described by the Langevin equation.[1]

Although in static or slowly flowing ferrofluids, the time required for magnetization to reach equilibrium is often neglected. This implies that the magnetization is always parallel to the magnetic field intensity. However, in reality, when the direction of the applied external magnetic field changes, the particles are subjected to a magnetic torque.[2] This torque tends to align the magnetic moments of the particles with the magnetic field direction, causing the magnetization of the ferrofluid to require a finite amount of time to reestablish equilibrium. Two primary mechanisms dominate this equilibrium process, depending on the particle size. One is the Brownian relaxation process,[9] which is realized through the physical rotation of the particles. In this case, the magnetic moment is fixed relative to the particle, and the rotation is influenced by the viscous resistance torque from the surrounding carrier liquid.[10] The other is the Néel relaxation process, where the magnetic moment inside the particles rotates relative to the crystal axis toward the magnetic field direction while the particle itself remains stationary.[11] These two relaxation processes are characterized by the Brownian relaxation time and the Néel relaxation time, respectively.

However, in flowing ferrofluids, the effect of magnetization relaxation becomes non-negligible when the magnetization relaxation time exceeds or is comparable to the characteristic flow time.[12] In such cases, under the influence of fluid viscous drag, the particles in the ferrofluids require a significant duration to reach equilibrium, or may fail

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to achieve an equilibrium state entirely. This manifests as the presence of a magnetization component in a direction other than that of the magnetic field at any given moment.[13] This effect induces flow characteristics in ferrofluids that differ from those of ordinary fluids. For example, the rotational viscosity[14] observed in ferrofluids under a static uniform magnetic field can inhibit Couette–Poiseuille flow[15,16] and alter its entrance length.[17] Meanwhile, static gradient and alternating magnetic fields can promote pipe flow[18] and channel flow[19–21] in ferrofluids due to the "negative viscosity" effect.[22] Furthermore, core rotation phenomena[23] can occur in ferrofluid pipe flows. Additionally, magnetization relaxation effects can lead to restricted cavity flow in ferrofluids and reduce the shear force acting on the driving plate.[24,25]

To calculate the off-field magnetization components resulting from magnetization relaxation in flowing ferrofluids, one approach is to numerically solve ferrohydrodynamic (FHD) models that incorporate magnetization equations.[26] It should be noted that the magnetization equation employed here is not the Langevin equation under equilibrium conditions. Instead, it is a dynamic equation that incorporates the effects of flow velocity and particle rotation. Examples include the Debye-like equation[27] or the Fokker–Planck equation within the mean field approximation.[28] Another approach employs molecular simulation methods, such as molecular dynamics and Langevin dynamics, to characterize microscopic structures within ferrofluids, such as particle chain formation. Subsequently, dipole-dipole interaction models, such as the modified mean field theory[29] and the Born-Mayer cluster expansion theory,[30] are utilized to establish the relationship between macroscopic equilibrium magnetic properties and the microstructure. Some researchers[31,32] have further introduced hydrodynamic interactions between particles into these microscopic methods. On the one hand, this inclusion serves to represent the vorticity diffusion that propagates the fluctuating motion of particles.[33] However, these studies still largely focus on the influence of such interactions on equilibrium magnetization[31,32] and are incapable of characterizing magnetization relaxation effects. On the other hand, by considering the influence of macroscopic shear flow in addition to hydrodynamic interactions, microscopic methods can be utilized to resolve the non-equilibrium magnetization of ferrofluids.[34–36]

In flowing ferrofluids, the description of the microelement stress state includes not only shear stress but also pressure. That is to say, in the study of magnetization relaxation effects, in addition to the macroscopic shear considered in existing microscopic methods, the influence of pressure should also be incorporated. Therefore, this study focuses on the component of magnetization perpendicular to the external magnetic field. Using a coupled simulation approach of molecular dynamics (MD) and Lattice Boltzmann (LB), the magnetization relaxation effects are explained from the perspective of how factors such as magnetic fields, shear, and pressure influence the microstructure within ferrofluids. Unlike continuum-based ferrohydrodynamics models, which rely on phenomenological magnetization equations, our coupled MD and LB method directly links the macroscopic non-equilibrium magnetization with the evolution of microscopic particle chain structures under the combined effects of magnetic fields, shear flow, and pressure gradients. This provides a mechanistic explanation for magnetization relaxation effects under complex flow conditions involving shear and pressure.

II. MODELS AND METHODS

A. Ferrofluid properties

Since the particles in ferrofluids are coated with surfactants, the particle size (diameter) can be expressed as

$$d_{\mathrm{p}} = x_{\mathrm{p}} + \delta, \tag{1}$$

where x_{p} represents the size (diameter) of the magnetic core, and δ is the thickness of the surfactant layer. Although the particle size in actual ferrofluids is generally polydisperse, previous research[37] has found that the bidisperse model yields results comparable to those of the polydisperse model when predicting magnetization. Therefore, the bidisperse model is adopted in this study.

Two important parameters influencing the microstructure of ferrofluids are the particle volume fraction and the dipole coupling coefficient. The latter represents the ratio of magnetic energy to thermal energy when the magnetic particles are in close contact. In a bidisperse ferrofluid with volume V, the volume fractions of the particles and the magnetic cores are given, respectively, by

$$\varphi = \frac{\pi N}{6}\frac{\langle d_{\mathrm{p}}^3\rangle}{V}, \tag{2}$$

$$\varphi_{\mathrm{m}} = \frac{\pi N}{6}\frac{\langle x_{\mathrm{p}}^3\rangle}{V}, \tag{3}$$

where N is the number of particles in that volume, the symbol $\langle\ldots\rangle$ denotes the average over the probability density distribution function of the magnetic core sizes of the particles, and its discrete form is used here, therefore,

$$\langle d_{\mathrm{p}}^3\rangle = d_{\mathrm{p},1}^3 \cdot \eta_{N,1} + d_{\mathrm{p},2}^3 \cdot \eta_{N,2}, \tag{4}$$

$$\langle x_{\mathrm{p}}^3\rangle = x_{\mathrm{p},1}^3 \cdot \eta_{N,1} + x_{\mathrm{p},2}^3 \cdot \eta_{N,2}. \tag{5}$$

Here, subscripts 1 and 2 denote the first and second particle sizes, respectively, $\eta_{N,1}$ and $\eta_{N,2}$ are the number proportions of particles corresponding to the first and second sizes, respectively.

The average dipolar coupling coefficient for a bidisperse ferrofluid is

$$\lambda = \frac{\mu_0}{4\pi}\frac{\langle m_{\mathrm{p}}^2\rangle}{k_{\mathrm{B}}T\langle d_{\mathrm{p}}^3\rangle}, \tag{6}$$

where μ_0 is the vacuum permeability, k_{B} is the Boltzmann constant, T is the temperature, and the average value of the particle's magnetic moment is

$$\langle m_{\mathrm{p}}^2\rangle = \left(\frac{\pi}{6}M_d\right)^2\left(x_{\mathrm{p},1}^6 \cdot \eta_{N,1} + x_{\mathrm{p},2}^6 \cdot \eta_{N,2}\right). \tag{7}$$

Here, M_d is the saturation magnetization of the particulate material.

B. Molecular dynamics model

For a ferrofluid system composed of N spherical particles dispersed in a cubic simulation box with a side length L, assuming that the particle magnetic moments are constant and located at the particle

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centers, the dipole–dipole interaction potential between the ith and jth particles is given by

$$u_{ij}^{DD} = \begin{cases} \infty & r_{ij} < d_{\mathrm{p},ij} \\ \dfrac{\mu_0 m_{\mathrm{p},i} m_{\mathrm{p},j}}{4\pi r_{ij}^3}\left[\boldsymbol{d}_i \cdot \boldsymbol{d}_j - 3\left(\boldsymbol{d}_i \cdot \hat{\boldsymbol{r}}_{ij}\right)\left(\boldsymbol{d}_j \cdot \hat{\boldsymbol{r}}_{ij}\right)\right] & r_{ij} \geq d_{\mathrm{p},ij}, \end{cases} \tag{8}$$

where $\boldsymbol{d}_i$ and $\boldsymbol{d}_j$ are the direction vectors of the magnetic moments $\boldsymbol{m}_{\mathrm{p},i}$ and $\boldsymbol{m}_{\mathrm{p},j}$, $\hat{\boldsymbol{r}}_{ij}$ is the unit vector pointing from particle i to particle j, r_{ij} is the magnitude of the vector from particle i to particle j, and $d_{\mathrm{p},ij} = (d_{\mathrm{p},i} + d_{\mathrm{p},j})/2$. The magnetic moment magnitude of one particular particle with a magnetic core diameter of $x_{\mathrm{p},i}$ is calculated as

$$m_{\mathrm{p},i} = \frac{\pi}{6} M_d x_{\mathrm{p},i}^3. \tag{9}$$

In addition to the dipolar potential, ferrofluid particles are also subjected to the short-range interaction potential. It is represented using a purely repulsive Lennard-Jones potential, expressed as

$$u_{ij}^{LJ} = 4\epsilon\left[\left(\frac{d_{\mathrm{p},i} + d_{\mathrm{p},j}}{2r_{ij}}\right)^{12} - \left(\frac{d_{\mathrm{p},i} + d_{\mathrm{p},j}}{2r_{ij}}\right)^{6}\right] + \epsilon, \quad r_{ij} < r_c, \tag{10}$$

where ϵ is the depth of the potential well, and the cutoff radius is chosen as

$$r_c = 2^{-\frac{6}{5}}\left(d_{\mathrm{p},i} + d_{\mathrm{p},j}\right). \tag{11}$$

This ensures a purely repulsive force that tapers off smoothly to zero at r_c.

Langevin dynamics is applied to simulate the motion of particles. In this case, the equations for the translational and rotational motion of the particles are given by

$$m_i \frac{\mathrm{d}\boldsymbol{v}_i}{\mathrm{d}t} = \boldsymbol{F}_i^{\mathrm{dip}} + \boldsymbol{F}_i^{\mathrm{LJ}} - \Gamma_{\mathrm{T},i}[\boldsymbol{v}_i - \boldsymbol{v}_c(\boldsymbol{r}_i)] + \boldsymbol{\xi}_{\mathrm{T},i}, \tag{12}$$

$$I_i \frac{\mathrm{d}\boldsymbol{\omega}_i}{\mathrm{d}t} = \boldsymbol{\tau}_i^{\mathrm{dip}} + \mu_0 \boldsymbol{m}_{\mathrm{p},i} \times \boldsymbol{H} - \Gamma_{\mathrm{R},i}\left[\boldsymbol{\omega}_i - \boldsymbol{\Omega}(\boldsymbol{r}_i)\right] + \boldsymbol{\xi}_{\mathrm{R},i}, \tag{13}$$

where $\boldsymbol{v}_i$ and $\boldsymbol{\omega}_i$ represent the translational and rotational velocities of the i-th particle, respectively, $\boldsymbol{r}_i$ is the position vector of the i-th particle, $\boldsymbol{v}_c(\boldsymbol{r}_i)$ and $\boldsymbol{\Omega}(\boldsymbol{r}_i)$ denote the translational velocity and rotational vorticity of the carrier fluid at position $\boldsymbol{r}_i$, respectively, m_i and I_i are the mass and moment of inertia of the i-th particle, respectively, and $\boldsymbol{H}$ is the magnetic field intensity. For spherical particles, their magnitudes are calculated by the following equations:

$$m_i = \frac{\pi}{6}\rho_{\mathrm{p}} d_{\mathrm{p},i}^3, \quad I_i = \frac{1}{10} m_i d_{\mathrm{p},i}^2, \tag{14}$$

where ρ_{p} is the density of the particle material. $\boldsymbol{F}_i^{\mathrm{dip}}$ and $\boldsymbol{F}_i^{\mathrm{LJ}}$ are the dipole interaction force and Lennard-Jones force acting on the particle, respectively, $\boldsymbol{\tau}_i^{\mathrm{dip}}$ is the dipole interaction torque acting on the particle, $\boldsymbol{\xi}_{\mathrm{T},i}$ and $\boldsymbol{\xi}_{\mathrm{R},i}$ represent the random force and random torque acting on the particle due to fluid thermal fluctuations, with mean values of zero. The terms $-\Gamma_{\mathrm{T},i}[\boldsymbol{v}_i - \boldsymbol{v}_c(\boldsymbol{r}_i)]$ and $-\Gamma_{\mathrm{R},i}\left[\boldsymbol{\omega}_i - \boldsymbol{\Omega}(\boldsymbol{r}_i)\right]$ represent the fluid drag force and drag torque exerted by the carrier fluid on the particle, respectively. $\Gamma_{\mathrm{T},i}$ and $\Gamma_{\mathrm{R},i}$ are the translational and rotational friction coefficients of the particle, which are given by

$$\Gamma_{\mathrm{T},i} = 3\pi\eta_{\mathrm{L}} d_{\mathrm{p},i}, \quad \Gamma_{\mathrm{R},i} = \pi\eta_{\mathrm{L}} d_{\mathrm{p},i}^3, \tag{15}$$

where η_{L} is the dynamic viscosity of the carrier fluid.

The formulation in Eq. (13), which rigidly fixes the magnetic moment to the particle, accounts for the Brownian relaxation mechanism. This is a valid assumption for the particle sizes (diameters $\gtrsim 10$ nm) and materials (magnetite) considered in this study, as the Brownian relaxation time is typically much shorter than the Néel relaxation time. The effects of Néel relaxation, which may become significant for very small particles (diameters $\lesssim 10$ nm), are beyond the scope of this work, which focuses on hydrodynamic coupling and microstructure.

C. Hydrodynamic model

The simulation box is treated as a fluid element in the flow field, which is subjected to fluid shear and pressure, as shown in Fig. 1. Assuming a velocity gradient exists in the x-direction for the ferrofluid element, the corresponding vorticity is $\boldsymbol{\Omega}$. According to the relationship between vorticity and flow velocity

$$\boldsymbol{\Omega} = \frac{1}{2}\nabla \times \boldsymbol{v}_c. \tag{16}$$

It follows that

$$\boldsymbol{\Omega} = -\frac{1}{2}\frac{\partial v_c}{\partial y}\boldsymbol{k}, \tag{17}$$

where the flow velocity $\boldsymbol{v}_c$ of the carrier fluid is along the y-direction, $\boldsymbol{k}$ is the unit vector along the z-direction. Assuming the shear rate in the carrier fluid is γ, we have

$$\gamma = \frac{\partial v_c}{\partial x}. \tag{18}$$

Consequently,

$$\gamma = -2\Omega. \tag{19}$$

Taking the flow velocity at the element face on the bottom plane yOz to be zero, the flow velocity in the linear shear flow is given by

$$v_c = \gamma x. \tag{20}$$

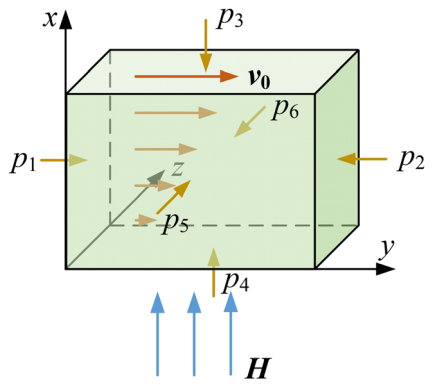


FIG. 1. Ferrofluid element and the forces acting on it.

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A uniform magnetic field in the x-direction is applied to the ferrofluid element. The magnetic field intensity is expressed as

$$\boldsymbol{H} = (H, 0, 0). \tag{21}$$

The LB method is employed to simulate the fluid influence. Assuming that in the fluid element shown in Fig. 1, the upper plane moves with velocity v_0 along the y-direction and the lower plane is stationary, it follows from Eq. (18) that the shear rate within the element is

$$\gamma = \frac{v_c}{L}. \tag{22}$$

In the macroscopic flow of a ferrofluid, a fluid element is subjected to pressure in addition to the viscous shear from adjacent fluids. Assuming the pressure difference acting on the element in a certain direction is Δp, and assuming the pressure on one side is zero, the dimensionless form of the pressure on the other side is

$$p = \frac{2P^* v_c \eta_{\mathrm{FF}}}{L^2}, \tag{23}$$

where η_{FF} is the dynamic viscosity of the ferrofluid and P^* is the dimensionless pressure gradient. In the y-direction, it can be expressed as

$$P^* = \frac{L^2}{2\eta_{\mathrm{FF}} v_c}\frac{\mathrm{d}p}{\mathrm{d}y}. \tag{24}$$

For the pressure boundary condition, according to the relationship between pressure p and carrier fluid density ρ_{L} in the LB method under isothermal conditions,

$$p = c_s^2 \rho_{\mathrm{L}}. \tag{25}$$

The pressure boundary condition is converted into a density value applied to the boundary nodes, where c_s is the lattice speed of sound. The dimensionless relationship corresponding to Eq. (25) is

$$p^* = c_s^{*2} \rho_{\mathrm{L}}^*, \tag{26}$$

where $c_s^* = \frac{1}{\sqrt{3}}$, thus yielding,

$$\rho_{\mathrm{L}}^* = 3p^*, \tag{27}$$

where ρ_{L}^* is the density value applied to the boundary nodes.

D. Scaling of the models

Taking the mean particle diameter x_0 and average mass m_0 as the length and mass references, and ϵ as the energy reference, all variables are expressed in the following dimensionless forms:

$$d_{\mathrm{p},i}^* = \frac{d_{\mathrm{p},i}}{x_0}, \quad x_i^* = \frac{x_i}{x_0}, \quad r_i^* = \frac{r_i}{x_0}, \quad r_{ij}^* = \frac{r_{ij}}{x_0}, \quad t^* = t\sqrt{\frac{\epsilon}{m_0 x_0^2}}, \tag{28}$$

$$T^* = \frac{k_{\mathrm{B}} T}{\epsilon}, \quad m_{\mathrm{p},i}^* = m_{\mathrm{p},i}\sqrt{\frac{\mu_0}{4\pi\epsilon x_0^3}}, \quad H^* = H\sqrt{\frac{4\pi\mu_0 x_0^3}{\epsilon}}, \tag{29}$$

$$m_i^* = \frac{m_i}{m_0}, \quad I_i^* = \frac{I_i}{m_0 x_0^2}, \quad \rho_{\mathrm{p}}^* = \rho_{\mathrm{p}}\frac{x_0^3}{m_0}, \quad \eta^* = \eta\frac{x_0^2}{\sqrt{m_0\epsilon}}, \tag{30}$$

$$v_i^* = v_i\sqrt{\frac{m_0}{\epsilon}}, \quad V^* = V\sqrt{\frac{m_0}{\epsilon}}, \quad \gamma^* = \gamma\sqrt{\frac{m_0 x_0^2}{\epsilon}}, \quad p^* = p\frac{x_0^3}{\epsilon}, \quad \omega_i^* = \omega_i\sqrt{\frac{m_0 x_0^2}{\epsilon}}, \tag{31}$$

$$F_i^{\mathrm{dip},*} = \frac{F_i^{\mathrm{dip}} x_0}{\epsilon}, \quad F_i^{\mathrm{LJ},*} = \frac{F_i^{\mathrm{LJ}} x_0}{\epsilon}, \quad \tau_i^{\mathrm{dip},*} = \frac{\tau_i^{\mathrm{dip}}}{\epsilon}, \quad \xi_{\mathrm{T},i}^* = \frac{\xi_{\mathrm{T},i} x_0}{\epsilon}, \quad \xi_{\mathrm{R},i}^* = \frac{\xi_{\mathrm{R},i}}{\epsilon}, \tag{32}$$

$$\Gamma_{\mathrm{T},i}^* = \Gamma_{\mathrm{T},i}\sqrt{\frac{x_0^2}{m_0\epsilon}}, \quad \Gamma_{\mathrm{R},i}^* = \Gamma_{\mathrm{R},i}\sqrt{\frac{1}{m_0 x_0^2\epsilon}}, \tag{33}$$

where t is the time, and the decorative symbol $(*)$ is used to denote a variable which has been properly scaled to be of order unity. With these nondimensional parameters, the nondimensional interaction potentials can be expressed as

$$u_{ij}^{DD*} = \frac{u_{ij}^{DD}}{\epsilon} = \begin{cases} \infty & r_{ij}^* < d_{\mathrm{p},ij}^* \\ \dfrac{m_{\mathrm{p},i}^* m_{\mathrm{p},j}^*}{\left(r_{ij}^*\right)^3}\left[\boldsymbol{d}_i\cdot\boldsymbol{d}_j - 3\left(\boldsymbol{d}_i\cdot\hat{\boldsymbol{r}}_{ij}\right)\left(\boldsymbol{d}_j\cdot\hat{\boldsymbol{r}}_{ij}\right)\right] & r_{ij}^* \geq d_{\mathrm{p},ij}^*, \end{cases} \tag{34}$$

$$u_{ij}^{LJ*} = \frac{u_{ij}^{LJ}}{\epsilon} = 4\left[\left(\frac{d_{\mathrm{p},i}^* + d_{\mathrm{p},j}^*}{2r_{ij}^*}\right)^{12} - \left(\frac{d_{\mathrm{p},i}^* + d_{\mathrm{p},j}^*}{2r_{ij}^*}\right)^{6}\right] + 1, \quad r_{ij}^* < r_c^*, \tag{35}$$

where $d_{\mathrm{p},ij}^* = \left(d_{\mathrm{p},i}^* + d_{\mathrm{p},j}^*\right)/2$ and $r_c^* = 2^{-\frac{6}{5}}\left(d_{\mathrm{p},i}^* + d_{\mathrm{p},j}^*\right)$. In this study, when the dipolar potential energy between two adjacent particles falls below $-1.5\lambda_i k_{\mathrm{B}} T\left[2d_{\mathrm{p},i}^* d_{\mathrm{p},j}^*/\left(d_{\mathrm{p},i}^* + d_{\mathrm{p},j}^*\right)\right]^3$ (Ref. 34), the two particles are considered bonded. This criterion identifies particle pairs whose dipolar attraction energy is significantly greater than the thermal energy $k_{\mathrm{B}}T$, indicating a stable bond.

The dimensionless equations of motion for the particles become

$$m_i^*\frac{\mathrm{d}\boldsymbol{v}_i^*}{\mathrm{d}t^*} = \boldsymbol{F}_i^{\mathrm{dip},*} + \boldsymbol{F}_i^{\mathrm{LJ},*} - \Gamma_{\mathrm{T},i}^*\left[\boldsymbol{v}_i^* - \boldsymbol{V}^*\left(\boldsymbol{r}_i^*\right)\right] + \boldsymbol{\xi}_{\mathrm{T},i}^*, \tag{36}$$

$$I_i^*\frac{\mathrm{d}\boldsymbol{\omega}_i}{\mathrm{d}t} = \boldsymbol{\tau}_i^{\mathrm{dip},*} + \boldsymbol{m}_{\mathrm{p},i}^* \times \boldsymbol{H}^* - \Gamma_{\mathrm{R},i}^*\left[\boldsymbol{\omega}_i^* - \boldsymbol{\Omega}^*\left(\boldsymbol{r}_i^*\right)\right] + \boldsymbol{\xi}_{\mathrm{R},i}^*. \tag{37}$$

E. Calculation of magnetization

According to the definition, the magnitude of the magnetization in a ferrofluid is the sum of the components of magnetic moments along the direction of the external magnetic field per unit volume, expressed as

$$M = \frac{\sum_{i=1}^{N} m_{\mathrm{p},i}^H}{V}. \tag{38}$$

Its dimensionless form is

$$M^* = \frac{M}{M_S}, \tag{39}$$

where M_S is the saturation magnetization of the ferrofluid, representing the magnetization corresponding to the state where the magnetic moments of all particles are aligned with the external magnetic field direction, calculated as

$$M_S = \left(\sum_{i=1}^{N} m_{\mathrm{p},i}\right)\Big/ V. \tag{40}$$

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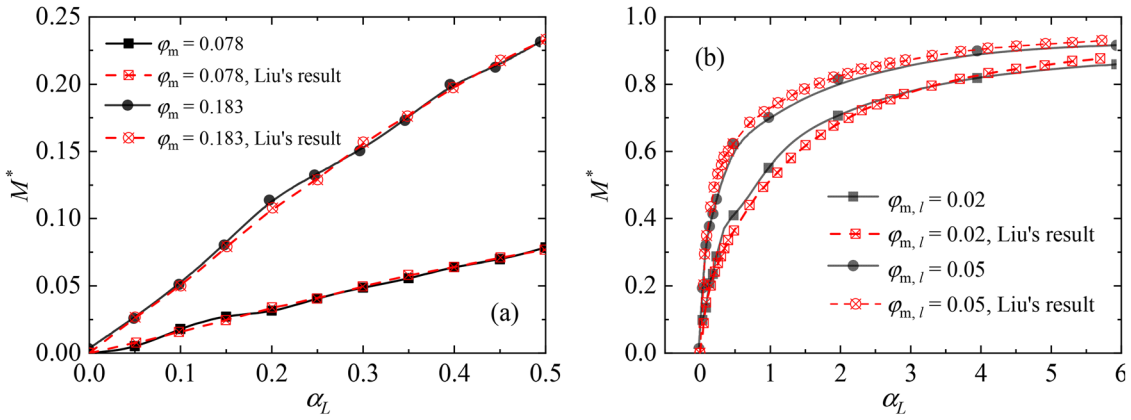


FIG. 2. Variation of dimensionless magnetization with the Langevin parameter for monodisperse (a) $\lambda = 3$ and bidisperse (b) $\lambda_s = 1.32$, $\lambda_l = 5.39$, $\varphi_m = 0.07$ systems.

Thus, the dimensionless form of the magnetization in the ferrofluid can also be written as

$$M^* = \left(\sum_{i=1}^{N} m_{p,i}^{H}\right) \Big/ \left(\sum_{i=1}^{N} m_{p,i}\right). \tag{41}$$

Utilizing the dimensionless magnetic moments from Eq. (29), the relationship (41) is further rewritten to

$$M^* = \left(\sum_{i=1}^{N} m_{p,i}^{H,*}\right) \Big/ \left(\sum_{i=1}^{N} m_{p,i}^{*}\right). \tag{42}$$

In this study, MD simulations of particle interactions in ferrofluids and hydrodynamic applications are performed using the Extensible Simulation Package for Research on Soft Matter (ESPResSo), which integrates both MD and LB modules. The side length of the cubic simulation box is given by

$$L = \left(\frac{\pi N x_0^3}{6\varphi_m}\right)^{\frac{1}{3}}. \tag{43}$$

Periodic boundary conditions are applied in all three directions, and the Ewald summation method is employed to calculate long-range dipolar interactions under metallic boundary conditions. Initially, the positions and orientations of the magnetic moments of all particles within the box are randomly determined.

III. MODEL VALIDATION

The validation of the models with coupled hydrodynamic interactions can be considered from aspects such as the equilibrium magnetization of a static ferrofluid, the velocity profile of the ordinary Couette–Poiseuille flow, and the non-equilibrium magnetization of a ferrofluid under pure shear.

The validation of the static ferrofluid magnetization is presented in Fig. 2, which illustrates the variation of equilibrium magnetization with the Langevin parameter for monodisperse and bidisperse systems, respectively. The correctness of the simulation method is validated by comparing it with data reported in the literature[34] for monodisperse systems with $\lambda = 3$, $\varphi_m = 0.078$ and 0.183, as well as for bidisperse systems with $\lambda_s = 1.32$, $\lambda_l = 5.39$, $\varphi_m = 0.07$ ($\varphi_{m,l} = 0.02$ and 0.05). Here, λ_s and λ_l represent the dipole coupling parameters for the small

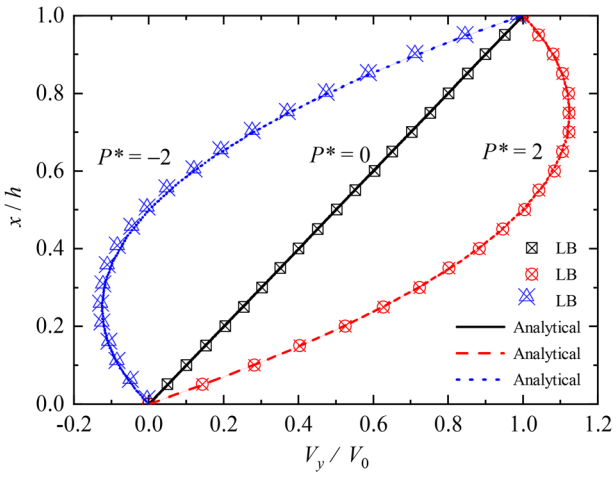


FIG. 3. Velocity profile of the Couette–Poiseuille flow for ordinary fluids.

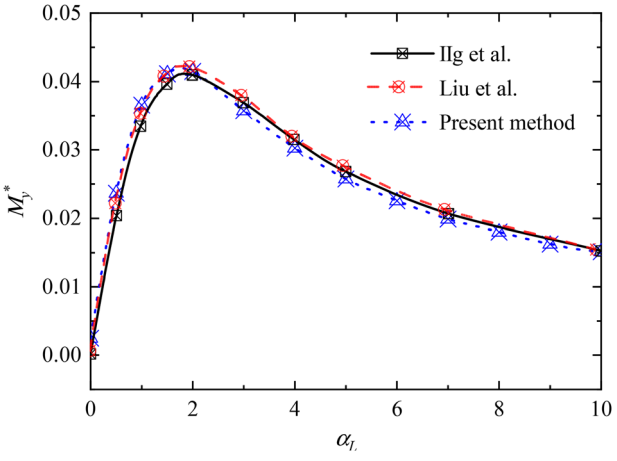


FIG. 4. Non-equilibrium magnetization of a ferrofluid under pure shear (monodisperse system, $\varphi_m = 0.1$, $\lambda = 1.0$, $\gamma^* = 0.1$, $P^* = 0$).

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and large particles in the bidisperse system, respectively, and $\varphi_{m,l}$ is the volume fraction of the large particles. It is evident that the results calculated using the method in this study show good agreement with the results of Liu *et al.*[34] In addition, the magnetization under static conditions was also validated in our previous research work[37] through comparison with experimental magnetization curves.

Regarding the Couette–Poiseuille flow of an ordinary fluid driven by a plate moving at velocity v_0, Fig. 3 compares the numerical results

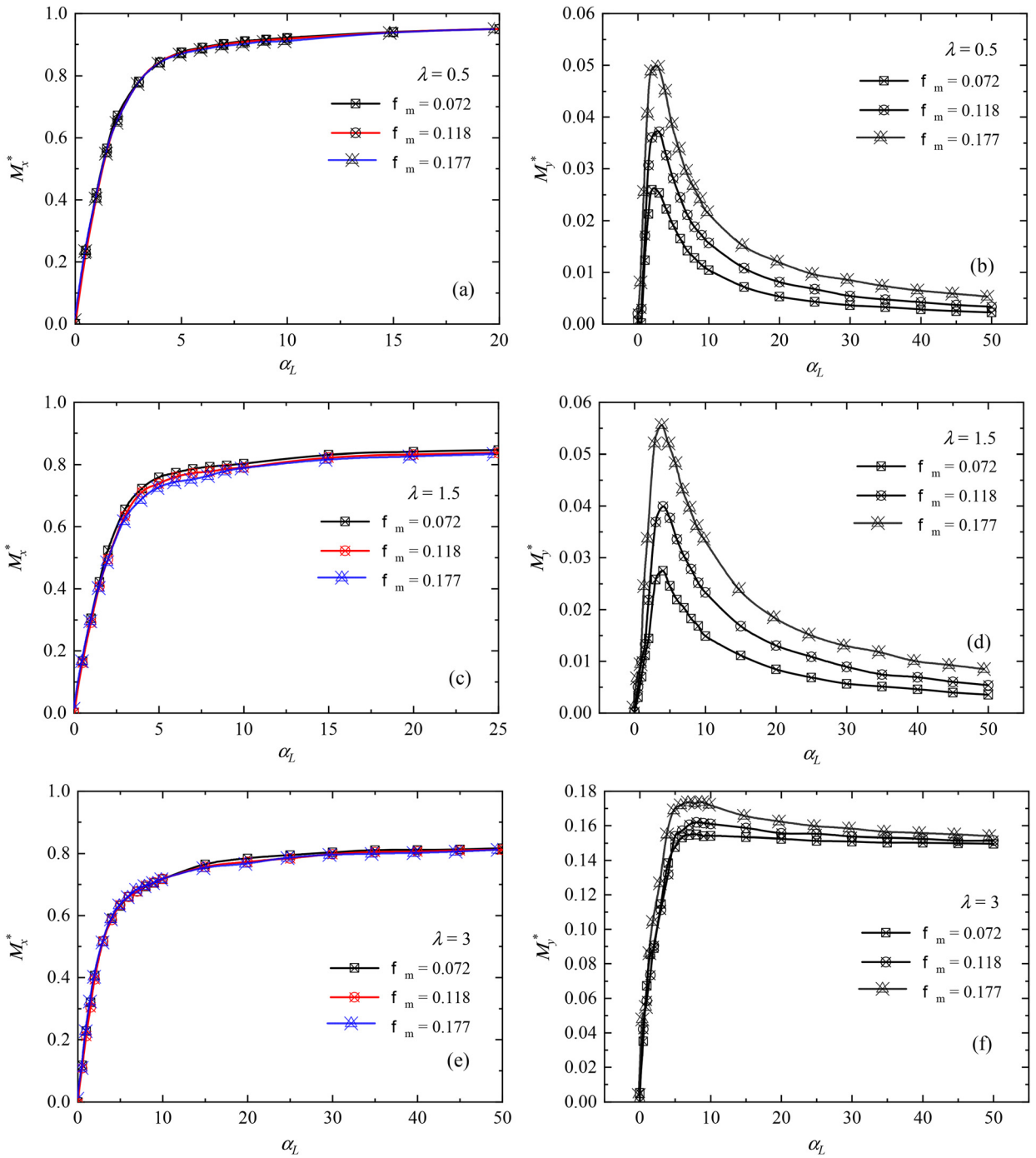


FIG. 5. Variation of the magnetization components of a ferrofluid with the Langevin parameter α_L for different concentrations and dipole coupling parameters ($\gamma^* = 0.1$, $P^* = 0.5$).

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obtained using the LB method of this model with the analytical solution. Excellent agreement is observed between the two, demonstrating the correctness of the LB calculation method employed in this study.

The model under non-equilibrium conditions is validated using the magnetization component perpendicular to the direction of the external magnetic field in a monodisperse system under pure shear flow. The magnetic field is applied along the x-direction, which is perpendicular to the flow direction. For a system with 3000 particles at $T^* = 1$, a time step of 0.002 is employed. The system is integrated for $t^* = 400$ to reach a steady state, followed by an additional sampling duration of $t^* = 500$. Figure 4 presents a comparison between the results calculated using the model in this study and those of Ilg *et al.*[38] and Liu *et al.*[34] under the same conditions. The three methods generally show good agreement, and slight differences may be attributed to different equilibration times.

IV. RESULTS AND DISCUSSION

Research with coupled hydrodynamic interactions is primarily used to predict the magnetization of ferrofluids. According to molecular dynamics simulation results from different dispersion models,[37] the bidisperse model can yield results almost identical to those of the polydisperse model and experimental values. Therefore, the bidisperse model is adopted in the subsequent parts of this study to simulate the magnetization characteristics of ferrofluids, as it accurately reproduces the magnetization characteristics of polydisperse ferrofluids while offering significant computational advantages.

Taking the commercial ferrofluid APG 513 A as an example, this study employs a bidisperse ferrofluid system in which the diameter ratio of the two types of particles is 1:1.5. Three values of the dipole coupling parameter λ (0.5, 1.5, and 3.0) are selected to represent systems with different interaction intensities, corresponding to average particle diameters of 10.5, 14.4, and 18.2 nm, respectively. Meanwhile, three particle volume fractions φ_m (0.072, 0.118, and 0.177) are chosen to represent dilute, moderately concentrated, and concentrated ferrofluids.

Simulations are performed at room temperature (300 K). As shown in Fig. 1, the system is subjected to shear flow $(0, \gamma x, 0)$ and a magnetic field $(H, 0, 0)$. The total number of particles in the system is fixed at 3000, and the temperature is set to $T^* = 1$. The dimensionless time step is set to 0.001. The system is integrated for a dimensionless time of $t^* = 1000$ to reach a steady state, and sampling is performed for an additional dimensionless time of $t^* = 1000$. The average of the sampled values is taken as the steady-state magnetization result. The reported values are time averages from a single long simulation for each parameter set, and their fluctuations reflect the inherent variability of the system. While ensemble averaging over multiple independent runs would provide more robust error estimates, the computational cost of coupled MD-LB simulations in our study necessitates minimizing statistical uncertainty through extended sampling durations.

A. Effects of particle concentration and size

Figure 5 shows the variation of the magnetization components of a ferrofluid with the Langevin parameter α_L at $\gamma^* = 0.11$ and $P^* = 0.5$. It can be observed that the magnetization component along the magnetic field direction, M_x^*, increases with increasing α_L and eventually reaches saturation, which is consistent with the magnetization behavior under equilibrium conditions. Under the influence of shear and pressure difference, the magnetization component perpendicular to the magnetic field, M_y^*, first increases and then decreases with α_L. The non-monotonic trend of M_y^* vs α_L is a direct manifestation of magnetization relaxation, as the magnetization lags behind the changing magnetic field direction due to fluid viscous drag. For given values of φ_m and λ, when α_L is small, the rotation angle of particle magnetic moments toward the magnetic field is small, resulting in low values for both M_x^* and M_y^*. As α_L increases, both M_x^* and M_y^* initially increase. However, when α_L exceeds a certain value, more particle magnetic moments align with the magnetic field direction, causing M_x^* to increase until saturation, while M_y^* decreases. The magnitude of M_y^* reflects the magnetization relaxation effect. From a microscopic perspective, the chain structures (or large particles) formed within the ferrofluid under the magnetic field are the main cause of the magnetization relaxation effect. Figure 6 presents the variation of the average chain length within the ferrofluid with the Langevin parameter α_L for $\lambda = 0.5$ and $\varphi_m = 0.118$. It is evident that this variation is almost consistent with the trend of M_y^* vs α_L. In a weak field (α_L small), chains are short/rare, leading to small M_y^*. As α_L increases, chains grow longer, enhancing the relaxation effect and increasing M_y^*. In a strong field (α_L large), most particle magnetic moments are strongly aligned with the field, reducing the misalignment (relaxation) and thus causing M_y^* to decrease. Therefore, it can be inferred that for fixed shear rate, pressure difference, particle concentration, and dipole coupling parameter, the behavior of M_y^* initially increasing and then decreasing with α_L is primarily caused by the formation and subsequent alignment of chain structures within the ferrofluid at different magnetic field intensities.

It can also be observed from Fig. 5 that the larger the particle concentration φ_m or the dipole coupling parameter λ, the greater the maximum attainable value of M_y^*. This can also be explained by the microstructure of the particles within the ferrofluid. For example, when $\gamma^* = 0.1$, $P^* = 0.5$, and $\alpha_L = 5$, for the cases of $\lambda = 0.5$, the average chain lengths corresponding to $\varphi_m = 0.072$ and $\varphi_m = 0.177$ are 2.25 and 3.03, respectively, and the percentages of unchained particles are 98.30% and 92.22%, respectively. For the cases of $\lambda = 1.5$,

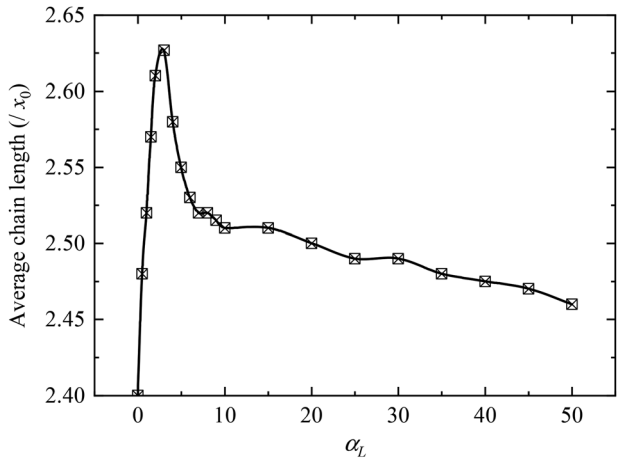


FIG. 6. Variation of the average chain length in a ferrofluid with the Langevin parameter α_L ($\lambda = 0.5$, $\gamma^* = 0.1$, $P^* = 0.5$, and $\varphi_m = 0.118$).

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the average chain lengths corresponding to $\varphi_m = 0.072$ and $\varphi_m = 0.177$ are 2.27 and 3.04, respectively, and the percentages of unchained particles are 98.05% and 91.29%, respectively. Similarly, when $\gamma^* = 0.1$, $P^* = 0.5$, and $\varphi_m = 0.118$, for the cases of $\alpha_L = 5$, the average chain lengths corresponding to $\lambda = 0.5$ and $\lambda = 1.5$ are 2.52 and 2.55, respectively, and the percentages of unchained particles are 95.95% and 95.63%, respectively. For the cases of $\alpha_L = 40$, the average chain lengths corresponding to $\lambda = 0.5$ and $\lambda = 1.5$ are 2.49

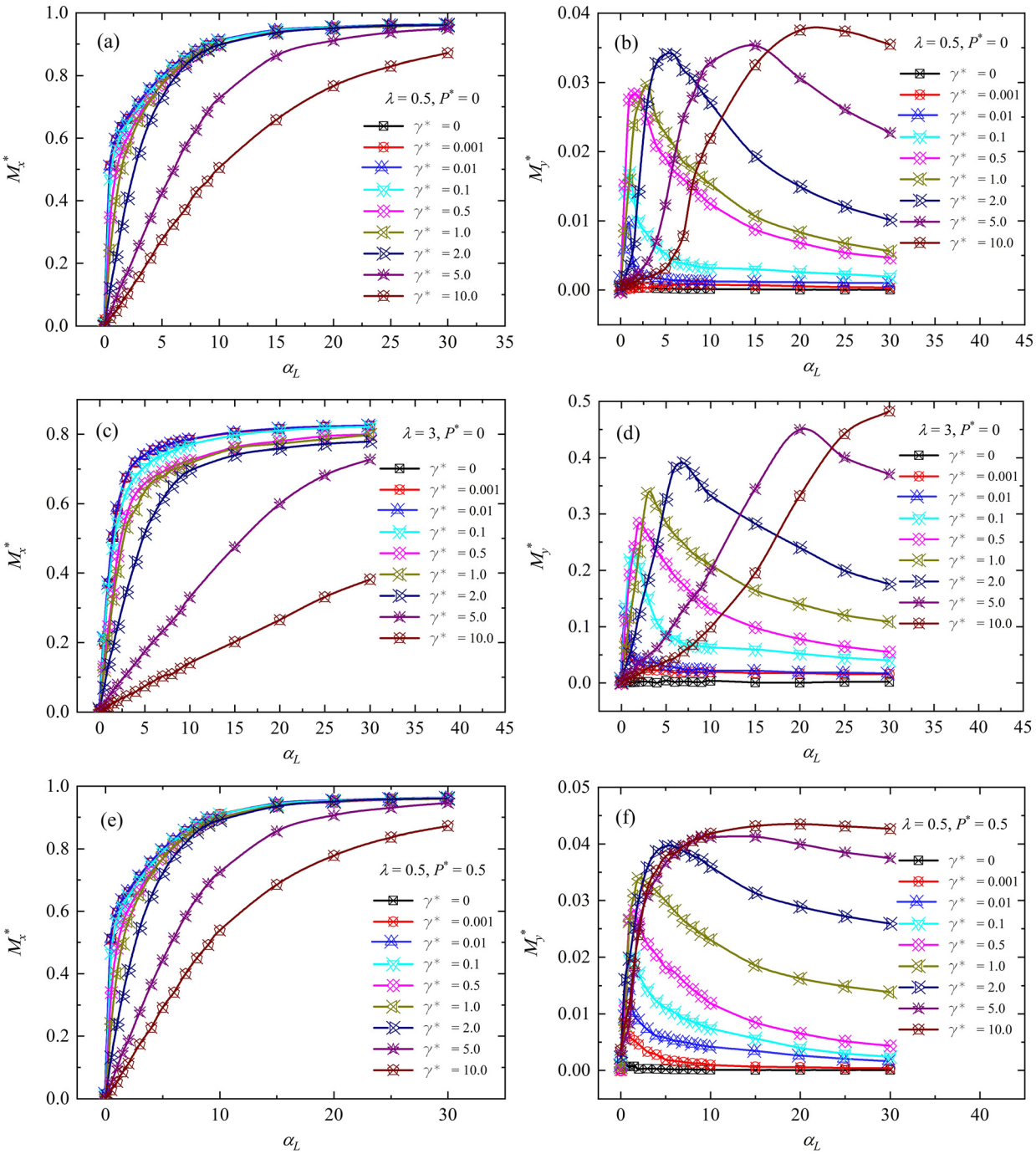


FIG. 7. Variation of the magnetization components of a ferrofluid with the Langevin parameter α_L at different shear rates ($\varphi_m = 0.118$).

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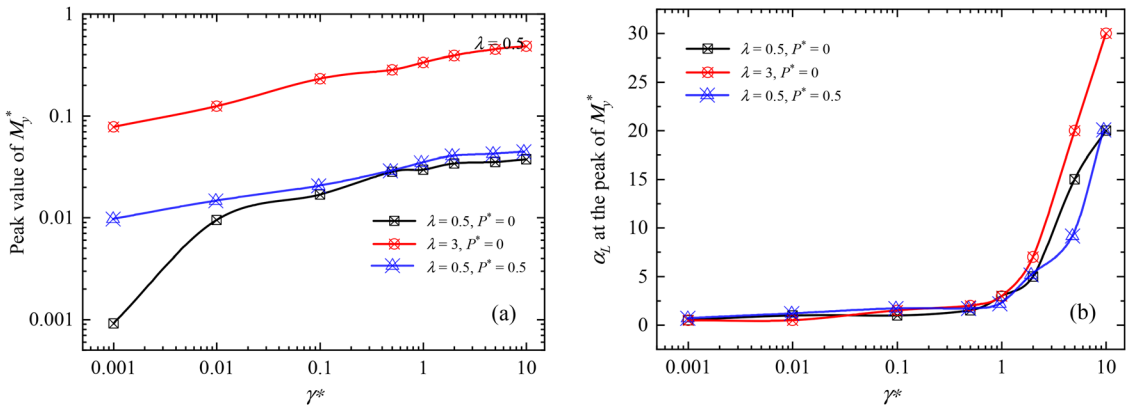


FIG. 8. Variation of the peak value of M_y^* (a) and the corresponding α_L (b) with shear rate γ^* ($\varphi_m = 0.118$).

and 2.50, respectively, and the percentages of unchained particles are 96.15% and 96.10%, respectively. In other words, the larger the particle concentration or the dipole coupling parameter, the longer the average chain length in the ferrofluid, and correspondingly, the larger the attainable value of M_y^*.

B. Effects of shear rate

Figure 7 shows the variation of the magnetization components of a ferrofluid with the Langevin parameter α_L at different shear rates, for $\lambda = 0.5$ and $\lambda = 3$, as well as $P^* = 0$ and $P^* = 0.5$. It can be seen that the larger the shear rate γ^*, the slower the magnetization component M_x^* in the magnetic field direction reaches saturation with increasing α_L, and this phenomenon is particularly pronounced for larger values of λ. For example, for the case of $\lambda = 0.5$ and $P^* = 0$, when $\alpha_L = 30$, M_x^* reaches 87.1% of its saturation value, whereas for the case of $\lambda = 0.5$ and $P^* = 0$, when $\alpha_L = 30$, M_x^* reaches only 38.2% of its saturation value. This is because a larger λ value corresponds to a larger particle size, and under the same magnetic field intensity, larger particles are more difficult to rotate toward the direction of the external magnetic field compared to smaller particles.

As in Sec. IV A, the magnetization component perpendicular to the magnetic field direction, M_y^*, initially increases and then decreases with α_L. However, a larger shear rate γ^* requires a stronger magnetic field intensity for M_y^* to reach its peak. Furthermore, the larger the shear rate γ^*, the larger the peak value of M_y^*. This can be clearly observed from Fig. 8. This variation of M_y^* results, on one hand, from the competition between the magnetic field and the shear flow: the shear flow drives the particles to rotate clockwise, whereas the magnetic field tends to align the magnetic moments of the particles with the magnetic field direction. In the weak field regime, the orientation of particles is weakly constrained by the magnetic field. As the shear rate γ^* increases, the tendency of particles to rotate due to shear becomes stronger, resulting in a larger initial slope for the curve of M_y^* vs α_L. When the magnetic field is strong, the magnetic field constrains the particle orientation more effectively. In this case, a flow with a larger shear rate is required to induce particle rotation. This leads to a decreasing trend in the M_y^* vs α_L curve at larger values of α_L. Additionally, for larger particles (i.e., larger λ values), a larger γ^* is required to induce significant rotation. On the other hand, flows with different shear rates induce different size distributions of microscopic chains within the ferrofluid. Taking the magnetic field corresponding to $\alpha_L = 5$ as an example, Table I presents the average size of particle chains and the proportion of unchained particles for three shear rate values γ^* under different conditions of P^* and λ. It can be observed that a non-zero shear rate acts to break particle chains, resulting in a shorter average chain size compared to the case without shear, and a corresponding increase in the number of unchained particles. However, certain values of the pressure difference P^* can attenuate this effect to a certain extent.

C. Effects of pressure difference

Figure 9 shows the variation of the magnetization components of a ferrofluid with the Langevin parameter α_L at different pressure differences P^*, for shear rates of $\gamma^* = 0.5$ and $\gamma^* = 1.0$. It can be seen that the effect of the pressure difference on M_x^* is similar to that of the shear rate γ^*. A larger P^* results in a slower saturation of the magnetization component M_x^* in the magnetic field direction with increasing α_L. However, this effect becomes significant only at larger P^* values, such as when the P^* value exceeds 2.0 in Fig. 9. The magnetization component perpendicular to the magnetic field direction, M_y^*, also initially increases and then decreases with α_L. Moreover, overall, a larger

TABLE I. Average size of particle chains and fraction of unchained particles in ferrofluids ($\varphi_m = 0.118$, $\alpha_L = 5$).

P^*	λ	γ^*	Average length of particle chains ($/x_0$)	Proportion of unchained particles (%)
0	0.5	0	2.84	92.45
		0.5	2.74	93.02
		10	2.45	96.88
	3.0	0.001	4.05	83.55
		0.5	2.76	92.86
		10	2.55	95.61
0.5	0.5	0	2.90	95.50
		0.5	2.47	96.69
		10	2.74	96.13

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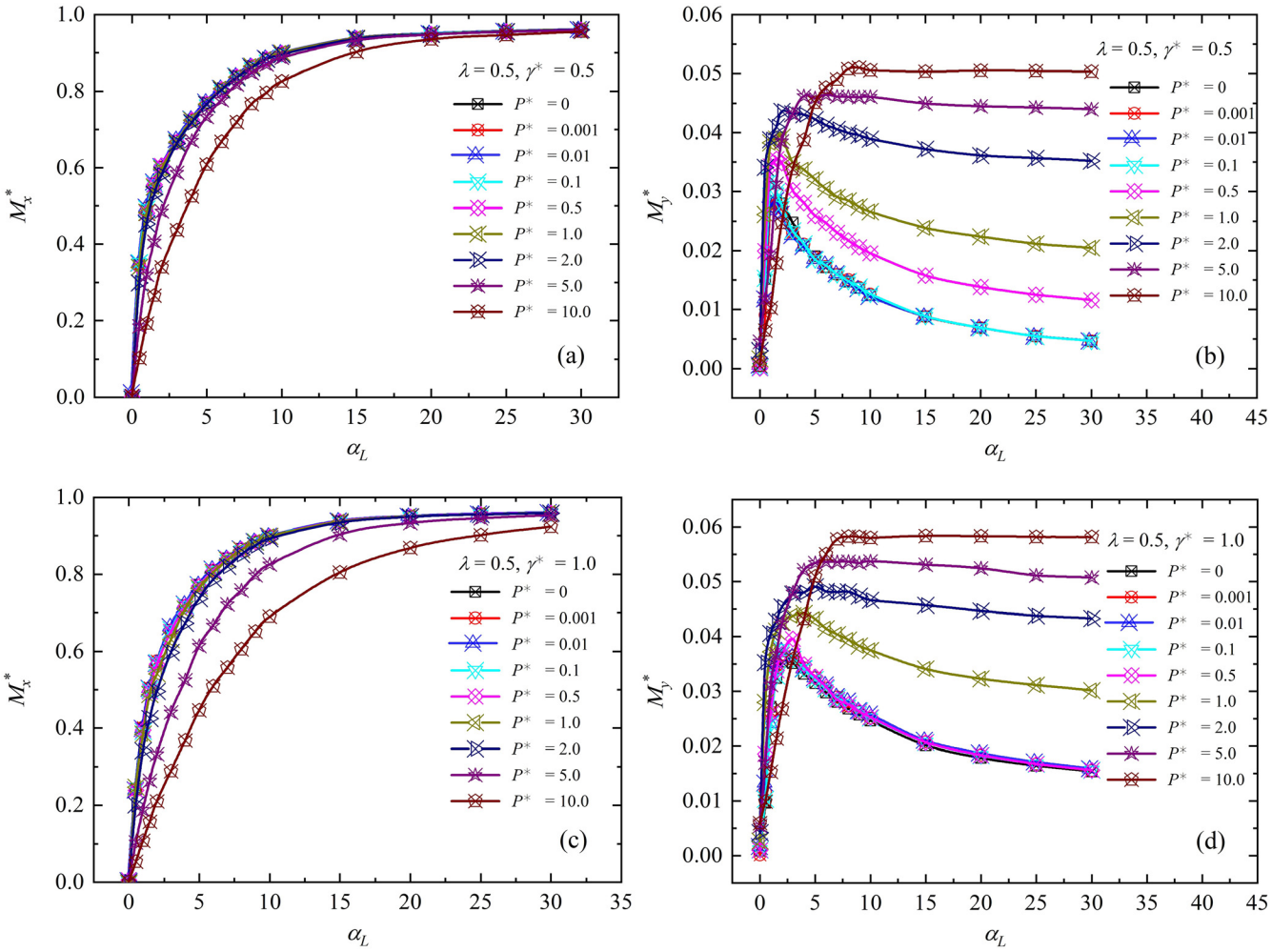


FIG. 9. Variation of the magnetization components of a ferrofluid with the Langevin parameter α_L at different pressure differences ($\varphi_m = 0.118$).

pressure difference P^* requires a stronger magnetic field intensity for M_y^* to reach its peak, and the larger the P^*, the larger the peak value of M_y^*. However, when the pressure difference P^* is small, both M_x^* and M_y^* hardly vary with P^*. Examples include the case of $P^* \leq 0.1$ at $\gamma^* = 0.5$, and the case of $P^* \leq 0.5$ at $\gamma^* = 1.0$. This is because when the pressure difference is small, the hydrodynamic effect mainly stems from the shear rate, and the hydrodynamic effect induced by the pressure difference becomes apparent only when the pressure difference exceeds a certain value. This can be seen from the variation of the local shear rate with P^* in the Couette–Poiseuille flow of ordinary fluids. Based on the velocity distribution of this flow pattern, the local shear rate can be calculated as

$$\frac{dv_y^*}{dx^*} = -2P^*x^* + P^* + 1. \tag{44}$$

The curve corresponding to this relationship is shown in Fig. 10. Since the P^* value corresponding to $\frac{dv_y^*}{dx^*} = 0$ is $P^* = 1.0$, it follows that when $P^* \leq 1.0$, the local shear rate of half of the fluid is less than that of the

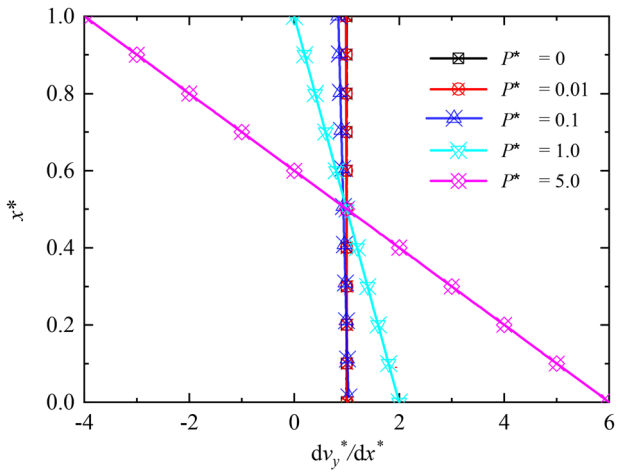


FIG. 10. Local shear rate in the Couette–Poiseuille flow of ordinary fluids.

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Couette flow ($P^* = 0$, which is constant at 1.0), while the other half is greater than 1.0. The net effect is that the influence of P^* cancels out within this P^* range. When $P^* > 1.0$, negative local shear rates appear within the fluid, and the absolute value of the local shear rate is greater than 1.0. At this point, the hydrodynamic interaction induced by P^* begins to manifest. Furthermore, the larger the P^* value, the greater the portion of the fluid where the absolute value of the local shear rate exceeds 1.0, and correspondingly, the larger the peak value of M_y^*. However, compared with ordinary fluids, in the Couette-Poiseuille flow of a ferrofluid, the uniform magnetic field perpendicular to the flow direction hinders the flow of the ferrofluid to a certain extent,[15] which causes a certain decrease in the critical P^* value at which the influence of the pressure difference on the magnetization begins to manifest.

V. CONCLUSIONS

In this study, in addition to the dipole interaction potential and Lennard-Jones potential between particles in ferrofluids, hydrodynamic interactions are coupled, and a bidisperse model of particle distribution is adopted. The variations in magnetization caused by these interactions are simulated using molecular dynamics and the Lattice-Boltzmann method. The magnetization component along the magnetic field direction, M_x^*, increases with the Langevin parameter α_L and eventually reaches saturation. However, larger particles are more difficult to rotate toward the external magnetic field direction than smaller particles under the same magnetic field intensity. Consequently, as the shear rate γ^* increases, or as the pressure difference P^* increases (once it reaches a certain value), the rate at which M_x^* reaches saturation slows down with increasing α_L. Under the influence of shear and pressure difference, the magnetization component perpendicular to the magnetic field direction, M_y^*, initially increases and then decreases with α_L due to differences in the chain structures within the ferrofluid at different magnetic field intensities. Moreover, the maximum attainable value of M_y^* increases with the particle concentration φ_m or the dipole coupling parameter λ. Due to the competition between the magnetic field and the shear flow, as well as the differences in the size distributions of microscopic chains induced by flows with different shear rates, a larger shear rate γ^* requires a stronger magnetic field intensity for M_y^* to reach its peak value. Furthermore, the larger the γ^*, the larger the peak value attained by M_y^*. In the studied flow configuration, the pressure difference P^* primarily influences magnetization by changing the local shear rate distribution within the fluid cell, rather than through a direct, independent microscopic coupling mechanism. The hydrodynamic interaction induced by the pressure difference becomes apparent only when P^* exceeds a certain threshold.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Wenming Yang: Conceptualization (lead); Funding acquisition (lead); Supervision (lead); Writing – original draft (equal). **Weitao Yuan:** Data curation (equal); Methodology (equal); Software (equal). **Xinhao Zhang:** Investigation (equal); Software (equal). **Beiying Liu:** Methodology (equal); Project administration (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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